aws STUDENT BUILDER GROUP CCOEWN

The Serverless side of AWS

A cost effective and scalable way of hosting your solutions

A bit about Me

- Born and brought up in Nagpur; graduated from VNIT Nagpur
- Software Engineer at Microsoft
- Previously at Fidelity Investments and GitHub
- Associated with various communities and colleges - as a speaker, mentor or as a judge!

ICEBREAKER

Kaju Katli vs Soan Papdi Battle



KAJU KATLI



SOAN PAPDI

https://dwyhq7u0dxeb3.cloudfront.net

Cloud Computing and AWS

- Rent compute, storage, and networking through APIs
- AWS provides these capabilities as individual services
- A Region determines where supported regional resources run

How Is Serverless Different?

- AWS operates the infrastructure behind the managed services
- Requests and events trigger application work
- We still own code, data, permissions, and cost controls

APIs, Routes, and Integrations

- An API exposes capabilities through network endpoints
- A route matches an HTTP method and path
- An integration connects the route to a backend such as Lambda



HTTP, REST, and WebSocket APIs

- HTTP APIs: a simpler option for many HTTP backends
- REST APIs: additional API-management capabilities
- WebSocket APIs: persistent, bidirectional client connections

Features at the API Boundary

- Authorization controls access to protected routes
- Throttling applies rate and burst settings
- CORS and access logs support browser integration and visibility

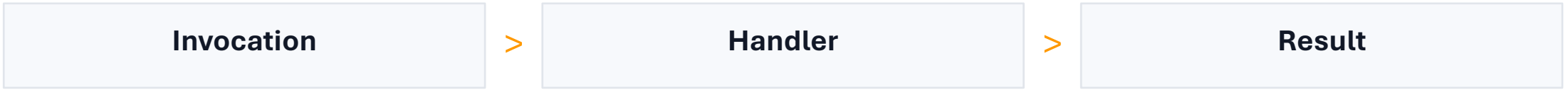
Follow a Request and Its Response

- POST /api/vote sends JSON to submit a choice
- GET /api/results reads the current round and counts
- Status codes distinguish success, invalid input, conflict, and overload

AWS LAMBDA

Functions, Handlers, and Events

- A function packages application code for managed execution
- The handler receives an event and invocation context
- The event shape depends on the invoking service



Runtimes and Function Configuration

- Choose a supported runtime and deployment package
- Memory influences compute resources; timeout bounds execution
- Environment configuration and an execution role support dependencies

Execution Environments and Cold Starts

- New environments initialize the runtime and application code
- Later invocations may reuse an existing environment
- Durable application state belongs outside function memory

Concurrency and Scaling Controls

- Concurrency is the number of invocations running at once
- Scaling depends on quotas and the capacity of downstream systems
- Reserved and provisioned concurrency solve different problems

Versions, Aliases, and Failure Handling

- Published versions provide immutable code/configuration snapshots
- Aliases point to versions and can support controlled traffic shifts
- Retry behavior depends on how the function is invoked

Tables, Items, and Attributes

- A managed NoSQL database for key-value and document data
- Tables contain items with typed scalar or nested attributes
- Non-key attributes can vary across items

Primary Keys and Access Patterns

- Use a partition key, or a partition key plus sort key
- GetItem targets one item; Query groups reads by key conditions
- Scan examines the table rather than targeting a primary key

Capacity and Read Consistency

- On-demand or provisioned capacity suits different usage patterns
- Eventual or strong consistency is available on supported read paths
- Hot keys, throughput limits, and throttling still matter

Atomic Updates, Conditions, and Transactions

- Atomic updates avoid lost concurrent counter changes
- Conditions allow a write only when an expected rule holds
- Transactions apply related checks and writes together or not at all

Active round > **Unique vote marker** > **Increment count**

Indexes, TTL, Backups, and Streams

- Secondary indexes support additional query patterns
- TTL expires records asynchronously
- Backups and Streams support recovery and change processing

WEBSITE DELIVERY

S3 Stores the Page; CloudFront Delivers It

- A private S3 bucket stores HTML, CSS, JavaScript, and photos
- CloudFront reads those objects through controlled origin access
- Students open the distribution's public HTTPS URL

WEBSITE DELIVERY

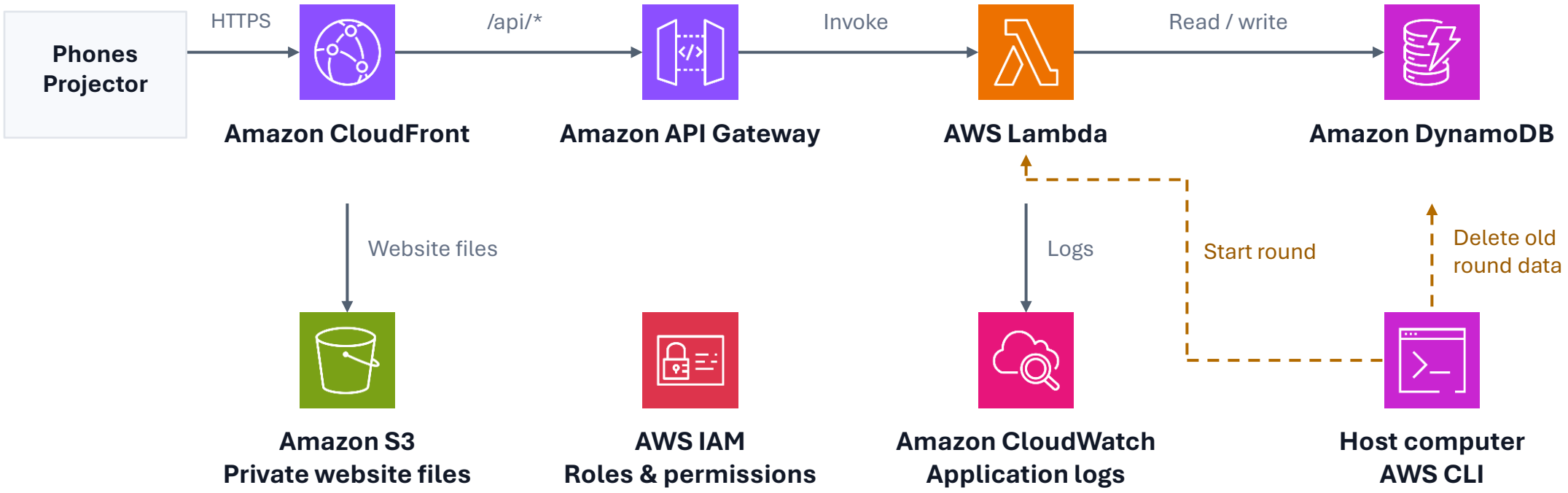
One Address, Two Origins

- Website paths route to S3
- /api/* routes to API Gateway
- Keep live results fresh with the appropriate cache policy

CloudFront > **S3 files / HTTP API**

SWEET SHOWDOWN ARCHITECTURE

Sweet Showdown: How the Components Connect



Solid: request / data paths Dashed: authenticated host actions

Thank you!

QUESTIONS & LIVE Q&A

